DelftX: MathMod1x Mathematical Modelling Basics

Help

Outline > 1. Introducing mathematical modelling > 1.5 Practice problem (Sunscreen lotion) > Validation

Validation

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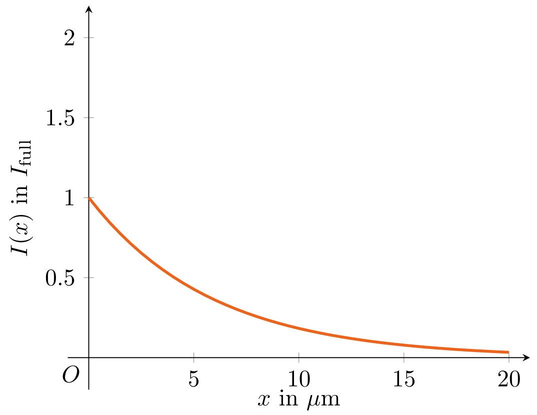
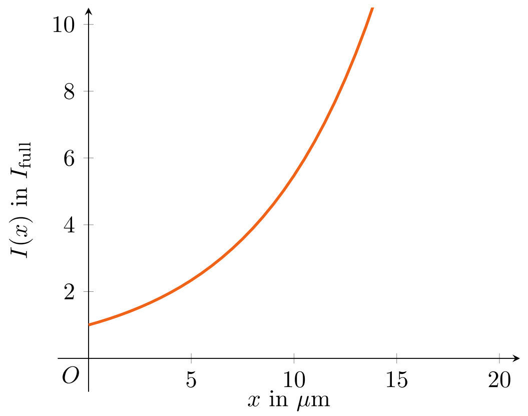
Time to validate the solution $I(x)$: Is it reasonable? Is it as expected? Are there measurements to compare to?

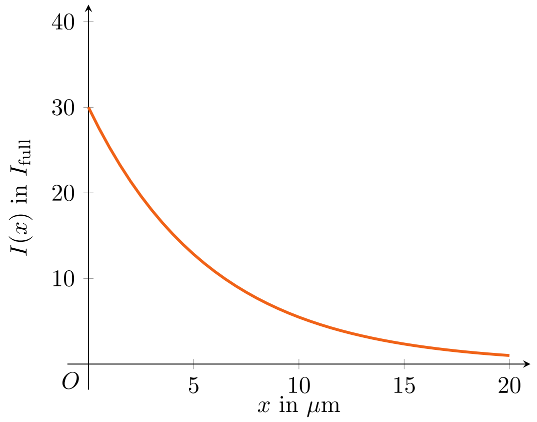
Exercise A

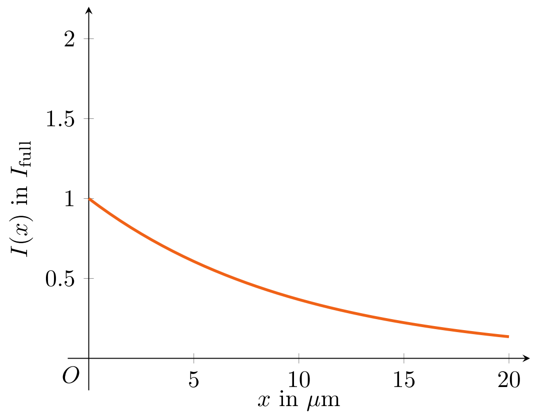
1/1 point (graded)

The graph of the light intensity in the 20μm layer of suncream with SPF=30 is









Answer
Correct: $I(x) = I_{full}\, e^{-0.17x}$

Submit You have used 1 of 1 attempt

Correct (1/1 point)

Exercise B

1/1 point (graded)
A layer of sunscreen lotion of SPF=30 with a thickness of 20μm should protect even the children with the lightest skins for 30 times 10 minutes in the full summer sun. How many minutes are the children protected when the layer is only 5μm thick?

Round your answer to full minutes.

23

Answer: 23

23

Submit You have used 2 of 3 attempts

Answers are displayed within the problem

Exercise C

1/1 point (graded)

Define $\mathrm{SPF}_{20\mu\mathrm{m}}$ as the Sun Protection Factor mentioned on the bottle of the sunscreen, which is calibrated for a layer thickness of 20 μm.

Define $\mathrm{SPF}_{5\mu\mathrm{m}}$ as the effective Sun Protection Factor when the suncream is applied in a layer of only 5 μm.

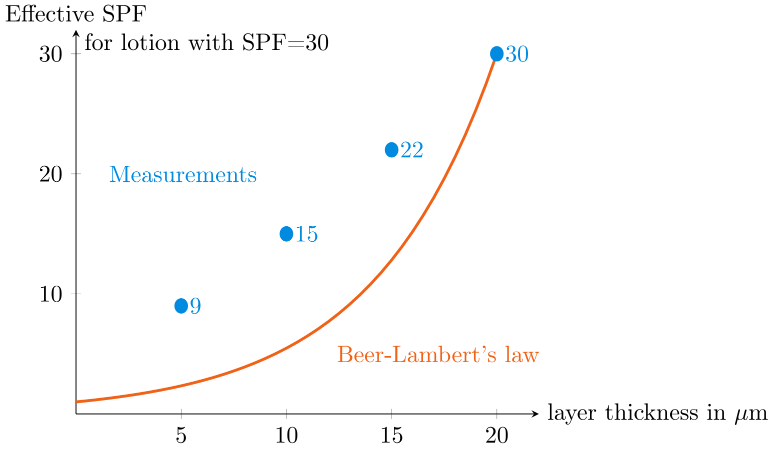
According to the model we derived, what is the relation between the two quantities?

- $\mathrm{SPF}_{5\mu\mathrm{m}} = \mathrm{SPF}_{20\mu\mathrm{m}} - 15$
- $\mathrm{SPF}_{5\mu\mathrm{m}} = \dfrac{\mathrm{SPF}_{20\mu\mathrm{m}}}{4}$
- $\mathrm{SPF}_{5\mu\mathrm{m}} = (\mathrm{SPF}_{20\mu\mathrm{m}})^4$
- $\mathrm{SPF}_{5\mu\mathrm{m}} = (\mathrm{SPF}_{20\mu\mathrm{m}})^{1/4}$ ✔

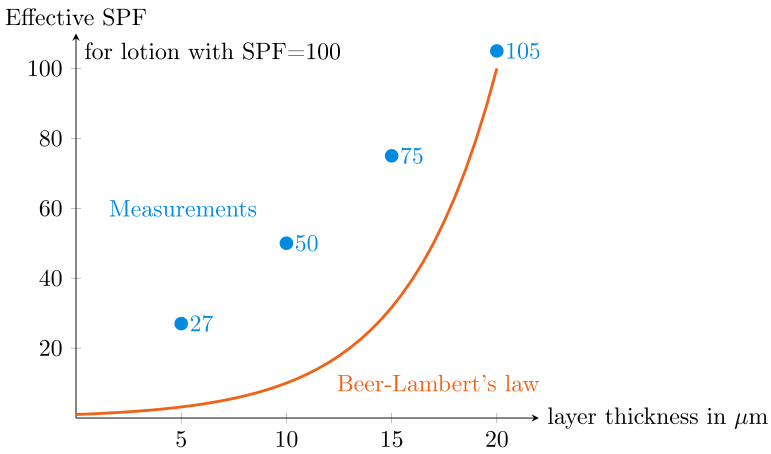
Submit You have used 1 of 1 attempt

Answers are displayed within the problem

The model we have derived for the transmitted light is called Beer-Lambert's law. According to this model, to protect the children from sunburn for 60 minutes on a sunny day, you would need sunscreen lotion with a SPF of nearly 1300! Luckily, that is not necessary. The *Effective SPF*, the protection factor for layers of different thicknesses has been measured (ofcourse not on children, but on healthy, paid volunteers). For a sunscreen lotion with a nominal SPF of 30, the measured values are given in the following graph.



For a sunscreen lotion with a nominal SPF of 100, the measured values are given in the following graph.



The discrepancy between the measurements and Beer-Lambert's law might be caused by any scattering and reflection that we ignored, but more probably because skin is not flat at all: a significant part of the suncream is used to fill the small crevices in the skin, and thus does not protect the high parts of the skin. The thickness of the 2 mg/cm^2 layer is probably much less for the high parts of the skin, increasing the effective SPF of the lotion.

Exercise D

1/1 point (graded)

So, the thicknesses of the layers can probably not easily be calculated from the mg of sunscreen lotion applied per cm^2. Let us go back to quantities: $\mathrm{SPF}_{2\mathrm{mg}/\mathrm{cm}^2}$, the effective Sun Protection Factor when the standard layer of 2 mg/cm^2 has been applied, and $\mathrm{SPF}_{0.5\mathrm{mg}/\mathrm{cm}^2}$, the effective Sun Protection Factor when a quarter of the same lotion has been used.

According to the measurements, what formula would best describe the relation between the quantities $\mathrm{SPF}_{2\mathrm{mg}/\mathrm{cm}^2}$ and $\mathrm{SPF}_{0.5\mathrm{mg}/\mathrm{cm}^2}$?

- [ ] $\mathbf{SPF}_{0.5\mathrm{mg}/\mathrm{cm}^2} \approx \mathbf{SPF}_{2\mathrm{mg}/\mathrm{cm}^2} - 15$

- (selected) $\text{SPF}_{0.5\text{mg/cm}^2} \approx \dfrac{\text{SPF}_{2\text{mg/cm}^2}}{4}$ ✔
- $\text{SPF}_{0.5\text{mg/cm}^2} \approx \left(\text{SPF}_{2\text{mg/cm}^2}\right)^4$
- $\text{SPF}_{0.5\text{mg/cm}^2} \approx \left(\text{SPF}_{2\text{mg/cm}^2}\right)^{1/4}$

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

Unexpected results, right? We'd like to hear your thoughts in the next discussion.